Explainable AI (XAI)

BMED365: Topic List X01–X08

BMED365

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Overview

1. **Why explainability?**
   - X01: Explain why explainability is important in medical AI
2. **Types of Explainability**
   - X02: Distinguish between global and local explainability
   - X03: Distinguish between ante-hoc and post-hoc explainability
3. **XAI Methods**
   - X04: Describe SHAP (SHapley Additive exPlanations)
   - X05: Describe LIME (Local Interpretable Model-agnostic Explanations)
   - X06: Explain Grad-CAM for CNN visualization
4. **Limitations and Challenges**
   - X07: Discuss limitations of XAI methods
   - X08: Know about attention visualization in LLMs

X01: Explain why explainability is important in medical AI

The challenge with "black box" AI:

- Complex models (deep networks, LLMs) give predictions without explaining *why*
- In medicine this is problematic for several reasons

Why explainability is critical in medicine:

1. **Trust:** Physicians and patients must trust the recommendations
2. **Accountability:** When errors occur – who is responsible?
3. **Regulation:** EU AI Act requires explainability for high-risk AI
4. **Learning:** Understand what the model has learned (and mis-learned)
5. **Debugging:** Identify bias and weaknesses
6. **Clinical integration:** AI as decision support, not autonomous

Medical practice

A physician who cannot explain why they recommend a treatment will lose the patient's trust. The same applies to AI.

X02: Distinguish between global and local explainability

Global explainability:

- Explains the model's **general behavior**
- "Which features are most important overall?"
- "How does age affect the prediction in general?"

Methods:

- Feature importance
- Partial dependence plots
- Global SHAP values

Local explainability:

- Explains one **specific prediction**
- "Why did *this* patient get a high risk score?"
- "Which pixels affected the classification?"

Methods:

- LIME
- SHAP (individual values)
- Grad-CAM (for images)

Both are important

Global: Understand the model as a whole, validate against domain knowledge
Local: Explain decisions to patients and colleagues

X03: Distinguish between ante-hoc and post-hoc explainability

Ante-hoc (built-in):

- Explainability **built into** the model
- The model is designed to be interpretable
- "Interpretable by design"

Examples:

- Linear regression (coefficients)
- Decision trees (rules)
- Attention weights (partially)

Advantage: True explanation
Disadvantage: Often less accurate

Post-hoc (after the fact):

- Explanation method **added after** training
- The model remains a "black box"
- Separate tools for interpretation

Examples:

- SHAP, LIME
- Grad-CAM
- Saliency maps

Advantage: Can be used on complex models
Disadvantage: Approximates, can be misleading

Trade-off

Often a choice between interpretability (ante-hoc) and performance (complex model + post-hoc)

X04: Describe SHAP (SHapley Additive exPlanations)

SHAP: Based on game theory

- Shapley values: Fair distribution of "payoff" among players
- Here: How much does each feature contribute to the prediction?

How SHAP works:

1. For each feature: Calculate average contribution to prediction
2. Considers all possible combinations of features
3. Gives a value per feature for each prediction

Strengths:

- Theoretically solid (Shapley axioms)
- Consistent and locally accurate
- Works for all model types

Visualizations:

- **Waterfall plot**: One prediction
- **Summary plot**: Global overview
- **Force plot**: Interactive

Python

```
import shap; explainer = shap.Explainer(model); shap_values = explainer(X)
```

X05: Describe LIME (Local Interpretable Model-agnostic Explanations)

LIME: Local approximation with simple model

Main idea:

1. Choose one prediction to explain
2. Generate **perturbed** (modified) versions of input
3. Run the complex model on all versions
4. Train a **simple, interpretable model** (e.g. linear) locally
5. Use the simple model to explain

Strengths:

- Model-agnostic
- Intuitively understandable
- Works for text, images, tables

Weaknesses:

- Unstable – different runs can give different answers
- Choice of neighborhood is arbitrary
- Can be misleading

Example: Text classification

LIME shows which words contributed most to the classification "spam" vs. "not spam"

X06: Explain Grad-CAM for CNN visualization

Grad-CAM: Gradient-weighted Class Activation Mapping

- Specialized for **convolutional neural networks (CNN)**
- Shows **which image regions** affected the classification

How it works:

1. Calculate gradients of output with respect to feature maps in last convolutional layer
2. Weight feature maps with average gradient
3. Combine into a **heatmap** that overlays the original image

Medical applications:

- X-ray/CT: "The model focused on this area for lung findings"
- Dermatology: "These pixels affected the melanoma diagnosis"
- Validation: Check that the model looks at the right area!

Important insight

Grad-CAM can reveal if the model uses **spurious correlations** (e.g. looks at text in the image, not pathology)

X07: Discuss limitations of XAI methods

XAI is not perfect – important limitations:

1. **Explanations are approximations**
   - Post-hoc methods don't explain the *true* model
   - Can be misleading or inconsistent
2. **Instability**
   - LIME can give different explanations for the same input
   - Small changes in input can cause large changes in explanation
3. **Interpretation burden**
   - Who should interpret the explanations? Requires expertise
   - Risk of over-reliance on explanations
4. **Computational cost**
   - SHAP can be very slow for large models

Remember

XAI is a tool for insight, not a guarantee that the model is "correct" or fair.

X08: Know about attention visualization in LLMs

Attention visualization in LLMs: Built-in "explanation"?

- Transformer models use **self-attention**
- Attention weights show which tokens the model "looks at"
- Can be visualized as heatmaps

Example:

*"The patient has **diabetes** and takes **metformin**"*

Attention visualization can show that "metformin" has high attention on "diabetes"

Opportunities:

- Insight into model focus
- Ante-hoc (built into the model)
- Easily accessible

Limitations:

- Attention $\neq$ explanation
- Many layers, many "heads"
- Can be difficult to interpret
- Shows correlation, not causation

Caution

Attention visualizations should be used as a supplement, not as a definitive explanation.

Summary: Explainable AI (XAI)

Key points:

- **X01:** Explainability is critical in medicine (trust, accountability, regulation)
- **X02:** Global vs. local explainability
- **X03:** Ante-hoc (built-in) vs. post-hoc (after the fact)
- **X04:** SHAP – game-theoretic, consistent
- **X05:** LIME – local approximation, model-agnostic
- **X06:** Grad-CAM – visualize CNN focus in images
- **X07:** Limitations – approximations, instability
- **X08:** Attention visualization in LLMs

Practical work in Lab 2 and Lab 3

- Lab 2: Use Grad-CAM to interpret CNN classification
- Lab 3: Discuss explainability and transparency in LLMs